

The cross-level double-edged-sword effect of boundary-spanning behavior on creativity

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Abstract

Boundary-spanning behavior has attracted considerable interest in recent years. Studies on this type of behavior have focused on its positive outcomes from the perspective of social networks. For decades, research has consistently demonstrated that the boundary-spanning behavior produces a wide array of positive results for teams and organizations. However, scholars have found that such behavior has negative outcomes for individuals. Using the conservation of resources theory (COR), we examined the double-edged-sword effect of boundary-spanning behavior on creativity at different levels, as well as its mediating mechanism and boundary conditions.

To test the proposed theoretical model, we applied multi-wave and multi-source research design. The data were collected from dyads of employees and supervisors in a company. At time 1, the boundary-spanning behavior, role stress, and role breadth self-efficacy were measured. These variables were rated by the employees. Approximately a month later, we asked the supervisors to rate the employees' creativity. These variables were assessed by mature scales. A total of 536 employees (90.32%) and 111 leaders (82.22%) responded to our survey. Confirmatory factor analyses and average variance extracted were conducted to assess the discriminant validity and convergence validity of the key variables. Multilevel structural equation modeling was used to validate the hypothesis, and Monte Carlo simulation procedures using open-source software R were conducted to test mediation effects.

Results showed that at the team level, boundary-spanning behavior had a significantly positive effect on team creativity ($\beta = 0.18$, $p < 0.05$). However, at the individual level, boundary-spanning behavior had a significantly negative effect on employees' creativity ($\beta = -0.02$, $p < 0.05$). At the individual level, boundary-spanning behavior had a significantly positive effect on role stress ($\beta = 0.05$, $p < 0.01$) and role stress had a significantly negative effect on creativity ($\beta = -0.34$, $p < 0.001$). The mediation effect of role stress was significant ($\beta = -0.02$, $p < 0.05$, Monte Carlo = 20000, 95% CI = -0.03 , -0.006). This evidence would indicate that the boundary-spanning behavior had a negative effect on individual creativity via role stress. Results also showed that the product term between the boundary-spanning behavior and role-breadth self-efficacy was significant ($\beta = -0.08$, $p < 0.01$). The role-breadth self-efficacy moderated the relationship so that the mediating effect of role stress was stronger for employees with low role-breadth self-efficacy.

The study illustrated the double-edged-sword effect of boundary-spanning behavior on creativity at different levels as well as the mediating mechanism and boundary conditions regarding the negative effect of boundary-spanning behavior on individual creativity, thereby enriching the literature on boundary-spanning behavior. Furthermore, this study identified the boundary conditions of COR, which broadens the scope of research on this theory.

Key words boundary spanning behavior; creativity; role stress; role-breadth self-efficacy

1 Introduction

1.1 Questions

In the current context of China's efforts to push forward the new era of comprehensive deepening reform, enterprises are faced with a changing environment, which requires enterprises to pay close attention to the external environment at all times, gather information and resources from outside the company to deal with volatile situations and to win in the fierce competition. So companies are increasingly, and urgently, encouraging employees to engage in boundary-spanning behavior. Boundary-spanning behavior refers to the behavior of an individual taking information from outside the organization and transmit-

ting it to people inside the organization (Katz & Tushman, 1983). Boundary-spanning behavior includes two levels, team and individual, which have been studied respectively in the literature (Marrone, Tesluk, & Carson, 2007). However, team boundary-spanning behavior is based on the individual level first and then aggregate to the team level (Chan, 1998). The two levels of boundary-spanning behavior are interrelated, individual boundary-spanning behavior is the basis of team boundary-spanning behavior (Liu & Li, 2014). In addition, according to the concept of boundary-spanning behavior, boundary-spanning behavior involves both employees and teams and has an impact on both, with the employee being the perpetrator of the boundary-spanning behavior, the team is the

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receiver of boundary-spanning behavior. Does boundary-spanning behavior have the same effect on the perpetrator and the receiver of the act? Previous studies have not systematically answer this question yet. Therefore, it is of great significance to study the boundary-spanning behavior at both team and individual levels in order to fully understand the effects of boundary-spanning behavior on the outcome variables at different levels.

Related research shows that boundary-spanning behaviors at the team level can improve team creativity (Xu & Qu, 2014), but at the individual level, we still know very little about what and how employees' boundary-spanning behavior affects individual creativity. In many cases, boundary-spanning behavior is an out-of-role behavior beyond the duty of role norms, and implementing boundary-spanning behavior spends a lot of additional resources such as time and energy of employees (Xu, 2019). Even in roles that require employees to span boundaries, employees also expend a lot of time and energy paying close attention to the external environment and obtaining information that is useful to the team (Katz & Tushman, 1983). The literature on creativity suggests that adequate resources are necessary to enhance individual creativity (Amabile, 1988). So from a resource perspective, boundary-spanning behavior can hamper creativity. Although it has been recognized that there may be negative effects of boundary-spanning behavior (Marrone et al., 2007), no further research has been done to explore how the negative effects of boundary-spanning behavior arise? And under what conditions?

Based on the conservation of resources theory and previous studies (Hobfoll, 1989, 2001; Hobfoll, Halbesleben, Neveu, & Westman, 2018), this study constructed and empirically examined the effects of boundary-spanning behavior on creativity at both team and individual levels, and further explore how and under what conditions these effects occur. Conservation of resources theory (COR) suggests that the loss of individual resources can lead to stress that can adversely affect the individual (Hobfoll, 1989; Hobfoll, Halbesleben, Neveu, & Westman, 2018). According to the conservation of resources theory, we think that at the team level, boundary-spanning behavior can obtain resources for the team and promote team creativity, but at the individual level, employees' boundary-spanning behavior can consume their resources, leading to role pressure, thus reducing the individual creativity of the employee. In addition, the literature on conservation of resources theory suggests that self-efficacy influences the individual stress-producing process (Hobfoll, 2001; Hobfoll, Freedy, & Geller, 1990). Based on this, we further examined the moderating effect of role-breadth self-efficacy on the above relationships.

1.2 Boundary-spanning behavior and creativity

Hobfoll (1989) was the first to develop the conservation of resources theory. The theory holds that individuals always strive to acquire, preserve, and protect resources. Hobfoll (1989) defined resources as "the individual's characteristics, conditions, and energies, that make individual feel valuable or the means by which individual acquires them." These include material resources (e.g., plant), conditional resources (e.g., seniority, position), personal characteristics (e.g., self-efficacy, and IQ), and capability resources (e.g., time, and knowledge). These resources are valued by individuals because they help them achieve their goals.

The literature on creativity suggests that adequate resources

are necessary to enhance individual creativity (Amabile, 1988). Inadequate resources are important barriers to individual creativity (Amabile, 1988). Without sufficient resources (such as time and knowledge), individuals will not be able to discover problems, find new ways to solve them and find new ideas (de Jonge, Spoor, Sonnentag, Dormann, & Van Den Tooren, 2012). According to the conservation of resources theory, the acquisition, preservation, and conservation of resources are essential for the survival and development of individuals (Hobfoll, 1989; Hobfoll et al., 2018). Not only the individual but also the team, in order to maintain their own survival and development, will continue to access resources. Team boundary-spanning behavior is an important way for a team to acquire resources. At the team level, boundary-spanning behavior aims to achieve the team goals (Choi, 2002). In the process of interaction with the outside, employees will consciously acquire useful information and resources for the team. Often, these resources are scarce within the team, and they are heterogeneous and novel compared to resources within the team. And heterogeneous, innovative resources are necessary to enhance the team's creativity (Zhang & Ge, 2016). In addition, boundary-spanning behavioral teams are proactive in reaching out to the organization's top leaders for support and commitment (Ancona & Caldwell, 1992), which is important for enhancing the team's creativity because of the high risks associated with creative activities, so it's easy to fail (Zhu, Xu, & Zhang, 2018). Obtaining the support of the leader can free the team's worries, and make the team feel comfortable to engage in creative activities, thus improving the team's creativity. Empirical studies have shown that team's boundary-spanning behavior improves team creativity (Xu & Qu, 2014).

But at the individual level, first of all, both in-and out-of-role boundary-spanning behavior, boundary-spanning employees need to pay close attention to the external environment for a long time, from the external complex and rapidly changing environment to obtain useful information for the team, this information needs to be translated into knowledge that team members can understand and then passed on to team members (Katz & Tushman, 1983), a complex process that consumes a lot of time and effort. The empirical study shows that there is a significant positive correlation between employees' boundary-spanning behavior and resource loss (Xu, 2019). Second, both in-and out-of-role boundary-spanning behavior, as long as employees span boundary, means that employees need to maintain contact with a wide range of stakeholders (Ancona & Caldwell, 1992), deal with a wide variety of different types of people and complex interpersonal relationships. Too much external interpersonal activities will interfere with employees' attention, resulting in employees can not focus on creative work, which has a negative impact on individual creativity. Empirical studies have shown that boundary-spanning behavior reduces employees' engagement at work (Ramrajan, Bezrukova, Jehn, & Euwema, 2011). Finally, for out-of-role behavior, on the one hand, employees generally need to engage in out-of-office behavior (Paul, Scott, & Sarah, 2011), which usually takes up employees' downtime, leading to employees don't have enough time to recover from work, and the restorative experience is an important factor in boosting employee creativity (Eschleman, Madsen, Alarcon, & Barelka, 2014). On the other hand, employees need to balance their in-role tasks with their out-of-role behaviors, which takes a lot of time and

effort. Individual resources such as time and energy are limited, and the loss of individual resources by boundary-spanning behavior will lead to the reduction of individual resources spent in creative activities. This is not conducive to the promotion of individual creativity. In conclusion, we propose the following hypothesis:

Hypothesis 1: At the team level, team boundary-spanning behavior promotes team creativity. But at the individual level, employees' boundary-spanning behavior can reduce their individual creativity.

1.3 The mediating effect of role stress

Conservation of resources theory holds that both actual and potential resource losses trigger an individual's stress and that after stress, the individual takes steps to relieve it (Hobfoll, 1989; Hobfoll et al., 2018). According to this theory, we propose that employees' boundary-spanning behaviors consume their time, energy, and other resources, thus resulting in role stress and have a negative impact on individual creativity. Role stress is a stress that occurs when an individual fails to fulfill his or her role expectations effectively. It includes three aspects: role conflict, role ambiguity, and role overload (Kahn, Wolfe, Quinn, Snoek, & Rosenthal, 1964). Role conflict refers to an employee's inability to meet the conflicting demands or expectations of multiple roles (Kim, Murrmann, & Lee, 2009; Rizzo, House, & Lirtzman, 1970). Role ambiguity refers to an employee's lack of clarity or understanding of his or her role and inability to achieve clear role expectations (House & Rizzo, 1972; Rizzo et al., 1970). Role overload is when the expectations of a role exceed the limits of one's capacity (Singh, 1998).

Role stress tends to occur in boundary-spanning activities (Wang, Liu, & Liu, 2017). Both in- and out-of role require the employee to span boundary. As long as the employee implements the boundary-spanning behavior, the employee needs to contact with different external stakeholders, and need to play different roles in interacting with different external stakeholders. Each role has different, and sometimes even conflicting demands on the employee (Rakarajan et al., 2011), and the frequent boundary-spanning activity of the employee requires constant switching between different and possibly conflicting roles. On the one hand, this is easy to cause the role of conflict; on the other hand, the employees will have confusion and not know clearly their own positioning, resulting in role ambiguity. If boundary-spanning behavior is extra-role behavior, on the one hand, in addition to the role tasks within the team, boundary-spanning employees also need to take on multiple tasks such as getting information from outside, processing information, and passing information to others. This increases the workload of the employees, and consumes a lot of time, energy, and other resources, leading to role overload (Deng, Liu, & Mao, 2018). Empirical studies show that the extra-role helping behaviors that consume employees' time and energy tend to cause role stress (Chen, Chen, & Luo, 2020). On the other hand, the task work within the role and the boundary-spanning behavior outside the role may not be consistent with the requirements of the employees, which easily leads to role conflict.

According to the conservation of resources theory, measures will be taken to prevent further resource loss when an individual feels stress (Hobfoll, 1989; Hobfoll et al., 2018). A creative activity consumes a lot of employee's resources. When the boundary-spanning employees due to excessive consumption of resources, resulting in the employees do not have enough re-

sources to invest in creative activities, in order to prevent further loss of resources, the employees will reduce the investment in creative activities, choose relaxation to restore resources (de Jonge et al., 2012), and wait until resources are restored before you get back to work. Empirical studies have shown that role stress reduces employee's engagement (Wang, Han, & Yang, 2014). In addition, role stress is often accompanied by emotional exhaustion (Wang & Zhang, 2016). Employees in emotional exhaustion often lack the motivation to work (Wan, 2011) and are not motivated to engage in creative activities. Empirical studies have shown that role stress reduces employee creativity (Çekmecelioglu & Günsel, 2011). Based on the above analysis, we propose:

Hypothesis 2: Employees' boundary-spanning behavior reduces employees' individual creativity by increasing their role stress.

1.4 The moderating effect of role breadth self-efficacy

Conservation of resources theory holds that individuals with more resources are less susceptible to resource loss and identifies self-efficacy as an important individual trait resource (Hobfoll, 2001) and an important factor of maintaining the individual resource pool (Hobfoll, 2001, p. 349), influencing the process by which loss of resources leads to stress (Hobfoll, 2001). Role breadth self-efficacy is a kind of important self-efficacy, and it is closely related to boundary-spanning behavior and role stress. Therefore, we propose that role-breadth self-efficacy moderates the relationship between boundary-spanning behavior and role stress. Role-breadth self-efficacy refers to an employee's perception of his or her ability to perform tasks that are broader, more proactive, and beyond the call of duty (Hwang, Han, & Chiu, 2015). It is a form of self-efficacy that emphasizes the employee's confidence to perform broader out-of-role behaviors (Galperin, 2012).

Boundary-spanning behavior is a very challenging and complex activity (Marrone et al., 2007) and the implementation of boundary-spanning behavior consumes a great deal of time and effort. Role-breadth self-efficacy (Hobfoll, 2001), as an important personal trait resource, can help employees to resist the resource depletion caused by boundary-spanning behaviors. According to the conservation of resources theory (Hobfoll, 2001), employees with a high role-breadth self-efficacy have more resources. Those with more resources are less sensitive to resource loss and less prone to stress. Specifically, employees with high role-breadth self-efficacy were more confident in their ability to manage relationships with stakeholders than employees with low role-breadth self-efficacy (Marrone et al., 2007), and this self-confidence can help employees to reduce the resources consumption caused by boundary-spanning behavior, thereby reducing the negative effects of boundary-spanning behavior, thus reducing role stress. Employees with high role-breadth self-efficacy were more confident of their ability to perform tasks beyond their assigned duties (Liao & Liang, 2015) and more confident of performing extra-role behaviors (Galperin, 2012), they have a high ability to cope with setbacks (Hartog & Belschak, 2012), and they can cope with difficulties (Liao & Liang, 2015). Therefore, in the face of the challenges and complexities of boundary-spanning behavior, such as conflicting role demands, role ambiguity, and role overload, they have the confidence to play a variety of roles and to be able to address these issues. therefore bound-

ary-spanning behavior is less likely to cause role stress. Based on the above analysis, we propose the following assumptions:

Hypothesis 3a: Role breadth self-efficacy moderates the relationship between boundary-spanning behavior and role stress. The higher the role breadth self-efficacy of employees, the weaker the positive effect of boundary-spanning behavior on role stress.

Furthermore, when the role stress of employees is reduced, the initiative and enthusiasm of employees to engage in creative activities are increased. In particular, employees with high role-breadth self-efficacy are more motivated to be active and innovative (Parker, Williams, & Nick, 2006), and more actively engaged in creative activities. Combining Hypothesis 2 and Hypothesis 3a, we propose the following first stage moderated mediation model hypothesis:

Hypothesis 3B: Role breadth self-efficacy moderates the mediating effect of role stress on employees' boundary-spanning behavior and individual creativity. The higher the employee's role breadth self-efficacy, the weaker the mediation of the role stress is.

2 Method

2.1 Sample

The sample of this study comes from several enterprises in Beijing, Shenzhen and Shanghai, mainly involving in finance, consulting, Internet and other industries. The researchers first explained the purpose of the survey to companies' leaders and asked them to identify a list of subordinates and supervisors who would be able to participate in the survey. The questionnaires were distributed mainly in paper format, and a small number of questionnaires were sent to the subjects by e-mail.

In order to minimize the effect of common method bias, a multi-stage and multi-source approach was used to collect data. The data collection was conducted in two phases, one from the subordinate and one from the supervisor. In the first phase, 620 subordinates were invited to participate in the study. We asked subordinates to rate their boundary-spanning behavior, role-breadth self-efficacy, role stress and subordinates' basic personal information. A total of 590 employees completed the survey. The second phase of data collection was conducted about a month later, involving 135 supervisors. We asked supervisors to rate the creativity of subordinates who had completed the first phase of the questionnaire and the supervisors' basic personal information, and a total of 120 supervisors completed the survey.

After two rounds of research, we matched the subordinate and supervisor questionnaires according to a pre-determined number. After deleting the unpaired and incomplete questionnaires, 536 employees' questionnaires (effective recovery rate 90.32%) and 111 supervisors' questionnaires (effective recovery rate 82.22%) were obtained. The team size (that is, the number of subordinates evaluated by each supervisor) ranged from 3 to 8, with an average team size of 4.83. In the subordinate sample, the proportion of men and women was basically equal, and the proportion of men was 48.1%. In the educational background, 61% were undergraduates, 22.8% were junior college students, 8.2% were graduate students, and above 8% were secondary school students and below 52.6% of employees were married. The average age of subordinates was 29.97 ($SD = 4.49$), and the average length of service in the unit was 3.66 years ($SD = 3.84$). The proportion of men and women in the

supervisor sample was also roughly equal, with men accounting for 55 percent. In education, 65.8% were undergraduates, 16.2% were junior college students, 16.2% were postgraduates and above, and 1.8% were secondary school students and below. 81.1% of supervisors were married. The average age of the supervisors was 34.05 ($SD = 4.68$), and the average length of service in the unit was 5.85 years ($SD = 3.97$).

2.2 Measurement

Boundary-spanning behavior. Using a scale developed by Faraj and Yan (2009), employees were asked to rate the extent to which they had received information and resources from outside the team. Using a seven-point scale, 1 to 7 represented "very inconsistent" to "very consistent". A total of 4 measurement items, a typical measurement item is "I will contact with important people outside the team to obtain information and resources for the team". Following Marrone et al (2007), the average value of each employee's boundary-spanning behavior is regarded as the boundary-spanning behavior at the team level. The reason for averaging, rather than variance or maximum or minimum, is that the variance approach is suitable for exploring variables that make up a diverse team, such as age diversity. The maximum or minimum method is suitable for situations where the team outcome depends entirely on the individual in the team (Barrick, Stewart, Neubert, & Mount, 1998). In this study, team's boundary-spanning behavior is mainly concerned with the number of boundary-spanning behavior, not the diversity of boundary-spanning behavior, so from our theoretical focus point of view, it is not suitable for the use of variance method. The number of the team's boundary-spanning behavior is not decided by an individual's boundary-spanning behavior in the team, but decided by the team members as a whole, so the team's boundary-spanning behavior is not suitable to use the maximum or minimum method. In the same way, team creativity is aggregated using the mean method. Cronbach's α is 0.80 for employee boundary-spanning behavior and 0.79 for team's boundary-spanning behavior.

Role breadth self-efficacy. Using a scale developed by Parker (1998), there are seven measurement items, and a typical measuring item is "I can contact people outside the company (such as customers) to discuss the problem". Using a five-point scale, 1 to 5 stands for "strongly disagree" and "strongly agree". Cronbach's α is 0.73.

Role stress. The scale was developed by Li and Zhang (2009), a total of 13 items, 5 of which were reverse scoring items. Using a five-point scale, 1 to 5 stands for "strongly disagree" and "strongly agree". A typical measurement item is "I received conflicting requests from two or more people". Cronbach's α is 0.76.

Creativity. Use a scale developed by Zhou and George (2001), a total of 13 items. We asked supervisors to rate their subordinates' creativity, and a five-point scale ranging from 1 to 5 representing "strongly disagree" to "strongly agree". A typical measurement item is "He/she will propose new ways to achieve the work goal". Supervisor's evaluation of each subordinate as an individual level of individual creativity, the supervisor's evaluation of all subordinates as a team level of team creativity. Cronbach's α of individual creativity is 0.97 and Cronbach's α of team creativity is 0.98.

Control variable. Previous studies have shown that demographics variables impact boundary-spanning behavior and creativity (Marrone et al., 2007), so we controlled for gender (1

female, 2 male), age (in years), marriage (1 unmarried, 2 married), education (1 junior high and below, 2 technical secondary/high school, 3 junior college, 4 undergraduate, 5 graduate and above) and length of service (in years) at the individual level, and the gender, age, marriage, education, and length of service of the leader at the team level, the same encoding as the corresponding variables at the employee level. Furthermore, individual paradoxical mindset, team functions (1 finance, 2 sales, 3 manufacturing, 4 operations, 5 design, 6 management, 7 others, translated into 6 dummy variables) and team size (in terms of team size) may also affect employees' boundary-spanning behavior and creativity. Therefore, the paradoxical mindset of individuals, team function and team size are also considered as control variables to rule out other possible explanations for the relationship between variables. Paradox mindset was measured using a scale developed by Miron-Spektor, Ingram, Keller, Smith, and Lewis (2018). There were a total of 9 items, and a typical measuring item is "I can better understand problems when I deal with conflicting perspectives". Cronbach's α of the scale is 0.86. In addition, the nesting effect may exist in enterprises with different team sources. In order to eliminate the nesting effect, enterprises (27 enterprises in total, converted into 26 dummy variables) are also used as control variables.

2.3 Data analysis

The data were analyzed with SPSS 22.0 and MPLUS 7.4. First, the Average Variance Extracted (Ave) and confirmatory factor analysis (CFA) were used to test the convergent validity and discriminant validity. Second, common method bias was tested by Harman single-factor method and untested single-factor method. Then, r_{wg} , ICC (1) and ICC (2) are used to test the validity of data aggregation. Finally, Path analysis in the multilevel structural equation Modeling (MSEM) was used to test the hypothesis and to calculate the confidence intervals of the mediating effect by Monte Carlo Method (Preacher & Selig, 2012). The moderated mediation model was examined using Edwards and Lambert (2007) path analysis.

3 Results

3.1 Validity test

Confirmatory factor analysis (CFA) was used to compare the convergent validity and discriminant validity of the vari-

ables. Team boundary-spanning behavior and team creativity are variables at the team level, while employees' boundary-spanning behavior, role stress, role breadth self-efficacy, individual creativity and paradoxical mindset are variables at individual level, so we construct cross-level confirmatory factor analysis with Mplus to test discriminant validity among variables. As there are many measurement items of role stress, individual creativity, team creativity, role breadth self-efficacy and paradoxical mindset, according to Wu and Wen (2011), in this paper, the balance method is used for packaging the measurement items of these five variables, and then the item with the highest factor load and the item with the lowest factor load are packaged into a measurement item, and so on. After packaging, role stress, role breadth self-efficacy, individual creativity and team creativity all included 3 items, paradox mindset included 4 items. The cross-level confirmatory factor analysis results show that the seven-factor model can fit the data well as shown in Table 1. In the model, the standardized factor load of the seven variables (employee's boundary-spanning behavior, role stress, individual creativity, role breadth self-efficacy, paradoxical thinking, team boundary-spanning behavior, team creativity) was are all above the critical value of 0.5 except for one item of team boundary-spanning behavior was 0.42 and slightly less than 0.5. And the AVE values of the seven variables were 0.59, 0.64, 0.87, 0.47, 0.64, 0.62, 0.93 respectively, except that the AVE values of the role breadth self-efficacy were slightly lower than the critical value of 0.5, the Ave values of the other six variables are all larger than the critical value of 0.5 (Anderson & Gerbing, 1988; Hair, Black, Babin, Anderson, & Tatham, 2006), which shows that the seven variables in the model have certain convergent validity. As shown in table 1, 11 competition models were significantly worse than the seven-factor model, indicating that 7 variables had discriminant validity. The Ave values of each variable are larger than the square of the correlation coefficient between the two variables (Fornell & Larcker, 1981; Hair et al., 2006), further indicating that the five variables have good discriminant validity.

3.2 Common method bias

To reduce the effect of common method bias, data were collected at different time points and from different sources in the study design phase, according to the recommendations of Podsakoff, Mackenzie, Lee and Podsakoff (2003). And for different

Table 1
Results of confirmatory factor analysis

Model	χ^2	df	χ^2/df	RMSEA	CFI	TLI	$\Delta\chi^2$	Δdf
Seven-factor model	369.05	122	3.03	0.06	0.96	0.95		
Six-factor model 1	1310.83	126	10.40	0.13	0.80	0.76	941.78***	4
Six-factor model 2	905.27	126	7.18	0.11	0.87	0.84	536.22***	4
Six-factor model 3	897.31	126	7.12	0.11	0.87	0.84	528.26***	4
Six-factor model 4	1651.73	126	13.11	0.15	0.75	0.69	1282.68***	4
Six-factor model 5	661.23	123	5.38	0.09	0.91	0.89	292.18***	1
Five-factor model 1	1835.17	129	14.23	0.16	0.72	0.67	1466.12***	7
Five-factor model 2	1602.10	127	12.61	0.15	0.76	0.70	1233.05***	5
Four-factor model 1	3078.33	131	23.50	0.21	0.51	0.41	2709.28***	9
Four-factor model 2	2487.72	131	18.99	0.18	0.61	0.53	2118.67***	9
Three-factor model	3721.34	132	28.19	0.23	0.41	0.29	3352.29***	10
Two-factor model	4013.51	133	30.18	0.23	0.36	0.24	3644.46***	11

Note. Individual level $N = 536$, team level $N = 111$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

variables were used 5 points and 7 points, in addition, the role stress measurement also set up 5 reverse scoring items. In the data analysis phase, Harman single-factor method and untested single-factor method were used to test the effect of common method bias. The results of Harman single-factor analysis showed that the KMO of exploratory factor analysis was 0.92, and the Bartlett spherical test was significant ($p < 0.000$). The unrotated first factor explained only 33.04% of the total variation. The unmeasured single-method latent factor showed that the AVE value of the common method was only 0.35, far below the critical value of 0.5 (Dulac, Coyle-Shapiro, Henderson, & Wayne, 2008), thus it is concluded that common method bias has no significant effect on the relationship between variables in this study.

3.3 Data aggregation

This study used cross-level design, team boundary-spanning behavior and team creativity as team-level variables, but the measurement was done at the individual level, so we need to test the feasibility of aggregation of data collected at the individual level to team level. According to the suggestions of Yu, Fang and Ling (2004), the validity of data aggregation was tested by r_{wg} , ICC (1) and ICC (2). The results showed that the group's boundary-spanning behavior r_{wg} median was 0.92, ICC (1) = 0.33, ICC (2) = 0.7, group creativity r_{wg} median was 0.98, ICC (1) = 0.57, ICC (2) = 0.86, and met the criteria of r_{wg} median > 0.7 (Lebreton & Senter, 2007), ICC (1) > 0.05 and ICC (2) > 0.5 (James, 1982). This indicates that team members' scores have sufficient intra-group consistency and inter-group differences, and intergroup differences accounted for 47.05% and 65.33% of the variance in team boundary-spanning behavior and team creativity, respectively. Thus they could be aggregated to the team level for follow-up analysis.

Table 2
Mean, standard deviation and correlation coefficient of variables

Variable	Mean	SD	1	2	3	4	5	6	7	8	9
At the individual level											
1. gender - employee	1.48	0.50									
2. age-employee	29.97	4.49	0.02								
3. education level- employee	3.68	0.76	0.07	0.01							
4. marriage - employees	1.53	0.50	-0.01	0.58***	-0.07						
5. tenure- employee	3.66	3.84	0.02	0.56***	-0.20***	0.37***					
6. paradoxical mindset	3.57	0.61	-0.01	0.02	0.04	0.04	0.18***	(0.64)			
7. boundary-spanning behavior of employees	4.78	1.15	-0.10	0.10*	0.05	0.06	0.13**	0.20***	(0.59)		
8. role stress	2.60	0.43	0.01	0.02	0.03	0.02	0.11**	0.08	0.15***	(0.64)	
9. role-breadth self efficacy	3.81	0.52	-0.06	0.11*	0.12**	0.11**	0.19***	0.56***	0.19***	-0.01	(0.47)
10. individual creativity	3.54	0.71	0.02	0.05	0.00	0.04	0.16***	0.27***	-0.02	-0.15***	0.33*** (0.87)
Team level											
1. gender-leadership	1.55	0.50									
2. age-leadership	34.05	4.68	0.08								
3. education-leadership	3.97	0.63	0.09	0.12							
4. marriage-leadership	1.81	0.39	0.44***	0.34***	0.05						
5. tenure -leadership	5.81	3.97	0.07	0.30**	-0.07	0.31**					
6. team size	32.57	92.33	-0.07	0.08	-0.28**	0.12	0.28*				
7. team boundary-spanning behavior	4.63	0.68	0.05	-0.02	0.16	0.13	0.20*	-0.04	(0.62)		
8. team creativity	3.54	0.72	0.04	-0.15	0.03	-0.08	0.21*	0.07	0.33**	(0.93)	

Note. Individual level $N = 536$, team level $N = 111$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The numbers in diagonal brackets are AVE for variables. Because the team functions and the enterprise have been converted to dummy variables, these two variables are not included in the table.

3.4 Descriptive statistical analysis

The mean, standard deviation and correlation coefficient of the variables were as shown in table 2. At the individual level, there was a significant positive correlation between employees' boundary-spanning behavior and role stress ($r = 0.15$, $p < 0.001$), and a significant negative correlation between role stress and creativity ($r = -0.15$, $p < 0.001$). On the team level, there was a significant positive correlation between team-crossing behavior and team creativity ($r = 0.33$, $p < 0.001$).

3.5 Hypothesis testing

According to Preacher, Zyphur and Zhang (2010), Mplus 7.4 was used to construct a two-level path analysis to test the hypothesis. The results were shown in figure 1, after controlling for paradoxical mindset at the individual level, gender, age, marriage, education, and tenure, as well as team function, team size, enterprise, leader's gender, age, marriage, education level and tenure, team boundary-spanning behavior had a significant positive effect on team creativity ($\beta = 0.18$, $p < 0.05$). The overall effect of employees' boundary-spanning behavior on individual creativity was negative and significant ($\beta = -0.02$, $p < 0.05$), which indicated that employees' boundary-spanning behavior had a significant negative effect on individual creativity. Hypothesis 1 was supported.

The results of the mediating effect of role stress showed that the boundary-spanning behavior had a significant positive effect on role stress ($\beta = 0.05$, $p < 0.01$), and the role stress had a significant negative effect on individual creativity ($\beta = -0.34$, $p < 0.001$). The mediating effect between role stress and individual creativity was significant (mediating effect = -0.02 , $p < 0.05$), and the 95% confidence interval of mediating effect did not include 0 (-0.03 , -0.006), which was shown by the Monte Carlo method, it is suggested that role stress mediates the rela-

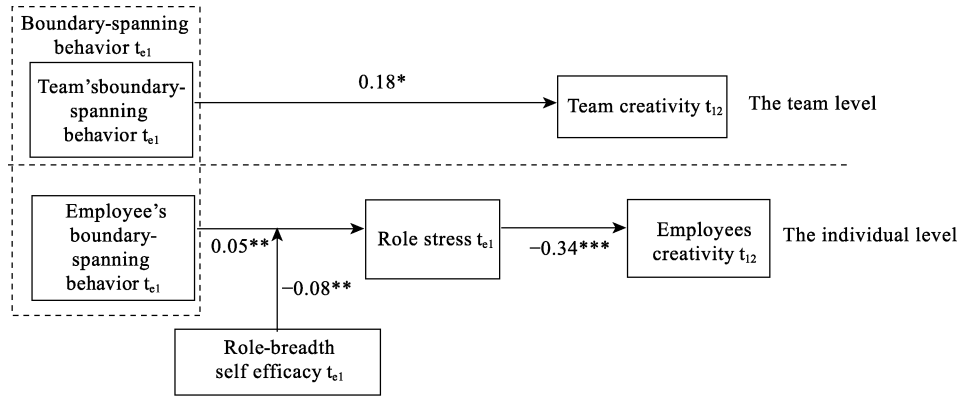


Figure 1. Results of the multilevel path analysis

Note. The path coefficient in the graph is the non-standardized path coefficient. T_{e1} indicates that the variable was evaluated by the employee at the first time point and t_{t2} indicates that the variable was evaluated by the leader at the second time point. Individual level $N = 536$, team level $N = 111$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

tionship between employees' boundary-spanning behavior and individual creativity. Hypothesis 2 is supported.

The results showed that the product item coefficient of boundary-spanning behavior and role-breadth self-efficacy was significant ($\beta = -0.08$, $p < 0.01$), it is suggested that role breadth self-efficacy moderates the relationship between boundary-spanning behavior and role stress. To clearly demonstrate the moderating effect of role breadth self-efficacy, we plotted the moderating effect graph (see figure 2) by adding or subtracting a standard deviation from the mean value of role breadth self-efficacy (Aiken and West, 1991). As shown in figure 2, when the employee's role breadth self-efficacy is low, the employee's boundary-spanning behavior has a stronger positive effect on role stress. The results of simple slope analysis showed that when role breadth self-efficacy was low (mean-one standard deviation), employees' boundary-spanning behavior had a significant positive effect on role stress ($\beta = 0.10$, $p < 0.001$). When the sense of self-efficacy of role breadth was high (mean + one standard deviation), the positive effect of boundary-spanning behavior on role stress was not significant ($\beta = 0.02$, $p > 0.1$), and the difference between the two groups was significant ($\beta = -0.08$, $p < 0.05$).

Following Edwards and Lambert (2007), we tested the moderated mediation model for the first stage of the hypothesis 3b. First, the samples were divided into high and low groups by adding or subtracting a standard deviation from the mean value of the moderation variables, and then the value and difference of the mediating effect of the two groups were calculated,

Table 3

The results of the moderated mediation model in the first stage

Group	Employee boundary-spanning behavior (X) $\rightarrow$ role stress (M) $\rightarrow$ individual creativity (Y)			
	Phase I (P_{MX})	Phase II (P_{YM})	Indirect effect ($P_{MX} \times P_{YM}$)	95% CI for indirect effect
High role breadth self-efficacy	0.02	-0.35**	-0.01	[-0.02, 0.01]
Low role breadth self-efficacy	0.10***	-0.35**	-0.03**	[-0.06, -0.01]
Intergroup difference	-0.08**	0	0.02*	[0.01, 0.05]

Note. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Since Mplus can not compute confidence intervals using bootstrap methods in cross-level analysis, we use the Monte Carlo method to compute confidence intervals in R software based on the recommendations of Preacher and Selig (2012) and their R language.

4 Discussion

From the perspective of resource, this study explores the effects of boundary-spanning behavior on team creativity and

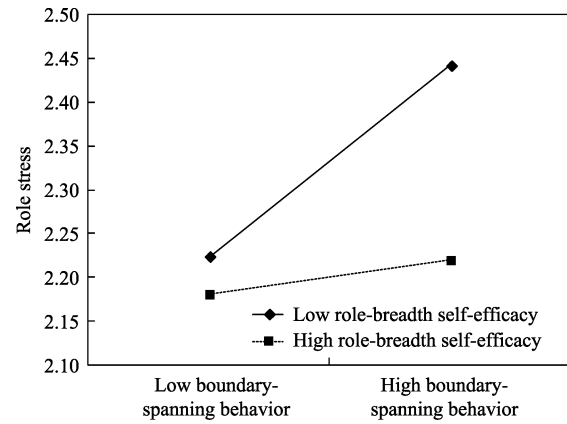


Figure 2. The moderating effect figure of role-breadth self-efficacy

respectively. Results are shown in Table 3. The mediating effect of role stress was small and not significant when the self-efficacy of role breadth was high (mediating effect = -0.01, $p > 0.1$, 95% CI including 0 [-0.02, 0.01]). When the self-efficacy of role breadth was low, the mediating effect of role stress was significant (mediating effect = -0.03, $p < 0.01$, 95% CI did not include 0 [-0.06, -0.01]), and there was significant difference between groups (the mediating effect value was 0.02, $p < 0.05$, 95% CI did not include 0 [0.01, 0.05]). It showed that role breadth self-efficacy moderated the mediating effect of role stress.

individual creativity, as well as the mediating mechanism and boundary conditions. The results show that team boundary-spanning behaviors improve team creativity at a team level, but at the individual level, employee boundary-spanning be-

haviors negatively affect individual creativity by increasing role stress. Compared with the employees with high role-breadth self-efficacy, the employees with low role-breadth self-efficacy were more likely to have role stress after implementing boundary-spanning behaviors. Thus the negative effect on individual creativity is even stronger. Scholars and practitioners should be alert to the adverse effects of boundary-spanning behavior and minimize their negative effects while giving full play to their positive effects.

4.1 Theoretical contributions

The theoretical contributions of this study are mainly reflected in the following three aspects: First, previous studies have either examined the effects of boundary-spanning behavior from the perspective of behavior receiver or perpetrator (Ancona & Caldwell, 1992; Song, Wang, & Zhang, 2017), this study integrates these two perspectives and explores the positive and negative effects of boundary-spanning behavior at both team and individual levels. More importantly, when focusing on the literature on the relationship between boundary-spanning behavior and creativity, we found that current research focuses on the impact of boundary-spanning behavior on behavioral receiver (team creativity) (Xu & Qu, 2014), this study focuses on the impact of boundary-spanning behavior on the behavior actor himself (individual creativity), that is, we shift the research perspective of the relationship between boundary-spanning behavior and creativity from the behavior receiver to the behavior actor himself, expanding from the team level to the individual level and helping scholars and practitioners to more fully understand boundary-spanning behavior.

Second, previous studies have focused on the positive effects of boundary-spanning behavior from a learning and network perspective (Liu, Jiang, Chen, Pan, & Lin, 2018). Empirical studies such as Xu and Qu (2014) showed that team boundary-spanning behavior improves team creativity through team learning. Liu and Li (2014) examined the mediating role of network-centeredness and found that employees' boundary-spanning behavior promoted their task performance by enhancing their network-centeredness. Different from previous studies, this study examines the negative impact of boundary-spanning behavior from the resource perspective. The results show that boundary-spanning behavior has a negative impact on individual creativity by increasing role stress. The results of this study show that in addition to the positive impact of boundary-spanning behavior, but also have a negative impact on employees, thus expanding the boundary-spanning behavior-related research.

Finally, the research enriches conservation of resources theory and finds the specific boundary condition of the resource preservation theory-The role-breadth self-efficacy. Halbesleben, Neveu, and Paustian Underdahl (2014) called people to pay attention to the interaction of multiple resources and emphasized the role of focusing on environmental factors. In response to the call of Halbesleben et al. (2014), the present study explored the moderating effect of the intrinsic psychological characteristic resources (role-breadth self-efficacy) on the process of resource depletion, which further enriched the theory of resource conservation.

4.2 Practical implications

Previous studies on boundary-spanning behavior and creativity have shown that boundary-spanning behavior can promote team creativity (Xu & Qu, 2014), but this study shows

that boundary-spanning behavior can not only promote team creativity, but also hinder individual creativity, and this suggests that the contribution of boundary-spanning behavior to team creativity may come at the expense of individual creativity. That is, the individual sacrifices the individual to achieve the greater good. Therefore, in practice, when teams enhance team creativity through members' boundary-spanning behaviors, we need to be alert to the negative effects of boundary-spanning behaviors on employees. On the one hand, we need to take measures to minimize the negative effects; on the other hand, we should encourage and reward the team spirit of the employees. For example, when building a team, try to choose employees with a high sense of self-efficacy. And more praise should be used to manage employees to improve their sense of self-efficacy. Because employees with a high sense of self-efficacy have less adverse effects when they engage in boundary-spanning behavior. In addition, team leaders can publicly praise and reward employees who engage in boundary-spanning behavior, and make other members feel that the implementation of boundary-spanning behavior is recognized and supported by the team. On the one hand, it can reduce the role stress and the negative influence of the boundary-spanning behavior. On the other hand, it is also an affirmation and recognition of the dedication of the team that the employees sacrificed themselves to achieve, thus encouraging the employee to make more efforts to contribute to the team.

4.3 Research limit and future research

First of all, although this study adopts a multi-stage and multi-source design to reduce the effect of common method bias to a certain extent, it still can not reach the conclusion of causality, in particular, boundary-spanning behavior and role stress were collected at the same time point. Future research may consider using the experimental way to test the relations between the variables, in order to obtain a reliable causal conclusion. In addition, the team creativity is measured by the way that team leaders evaluate each member's individual creativity, rather than by the team as a whole, which is also the shortcoming of this study, future research will allow team leaders to directly evaluate the overall creativity of the team, and further verify the conclusions of this paper.

Second, in many cases, boundary-spanning behavior may not be voluntary, but rather orchestrated by the team to adapt to the external environment (Ancona & Caldwell, 1992), so it can be divided into active and passive behaviors. The effects of these two different types of boundary-spanning behavior may be different. Future studies could continue to explore the positive and negative effects and mechanisms of different types of boundary-spanning behavior. In addition, this study only focused on the stress mechanism. At the individual level, in addition to role stress, there may be other mechanisms that lead to a positive impact on creativity. For example, if boundary-spanning behavior employees have rich sources of information, employees can easily access a wealth of even new ideas, which may promote individual creativity. Future research will continue to explore the dual and even multiple mechanisms by which boundary-spanning behaviors have both positive and negative effects on creativity at the individual level.

Finally, this study is conducted in China, and whether the conclusions can be extended to other countries remains to be further tested. China is a country with high collectivism (Hofstede, Hofstede, & Minkov, 2010), and Chinese employees

are more likely to sacrifice their own interests for the sake of the collective good, so they are more likely to take the initiative to span boundaries in order to improve team performance. In contrast, the United States is a country with a higher tendency to individualism (Hofstede et al., 2010). American workers are more concerned with their own interests and less likely to sacrifice their own for the collective good, so they are less likely to span boundaries on their own initiative. That is, employees from different cultural backgrounds may have different possibilities and motivations for boundary-spanning behavior. Future studies can collect data in different countries at the same time for comparative studies and may draw interesting conclusions.

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